

NourishAI: An Intelligent Personalized Nutrition and Diet Recommendation System

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1 Abstract

This project documents the development of NourishAI, an end-to-end artificial intelligence application designed to address the growing global need for personalized nutritional guidance. By integrating a Random Forest machine learning classifier with a responsive web interface, the system generates custom meal plans tailored to specific user biometrics, activity levels, and health goals. The application emphasizes clinical safety by incorporating health-aware filtering for conditions such as Thyroid and Diabetes. Our findings demonstrate that utilizing a multi-layered approach—combining predictive modeling with heuristic nutritional constraints—provides a robust framework for digital health interventions.

2 Introduction

Modern dietary management often suffers from a "one-size-fits-all" approach that fails to account for individual physiological variations and medical requirements. NourishAI was developed to bridge this gap by providing a hands-on implementation of the AI project lifecycle, from data acquisition to UI deployment. The system utilizes a dataset of over 400 recipes to match user profiles with optimal meal options. By calculating Total Daily Energy Expenditure (TDEE) and leveraging supervised learning to rank recipe compatibility, NourishAI provides users with a scientifically grounded roadmap for weight management, muscle gain, and overall wellness maintenance.

3 Methodology and System Architecture

The NourishAI system is built upon a modular architecture designed to bridge the gap between complex AI inference and a user-friendly interface. The methodology is divided into two distinct perspectives: high-level system interaction and low-level machine learning logic.

3.1 High-Level System Integration

The application follows a client-server paradigm where the frontend acts as a data collection and presentation layer, while the Flask backend serves as the orchestration engine.

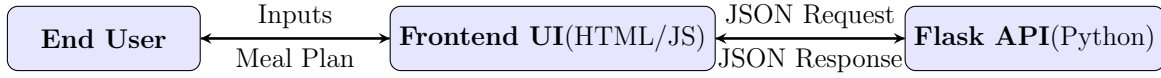


Figure 1: High-Level Request-Response Cycle

As shown in Figure 1, the process begins when the user submits biometrics through the UI. The `app.js` script performs metric conversions and dispatches an asynchronous POST request to the server. The server processes this request and returns a structured diet plan, which is dynamically rendered by the browser.

3.2 Low-Level AI Logic Pipeline

The core "intelligence" of NourishAI resides in its multi-stage processing pipeline, which combines heuristic nutritional science with supervised learning.

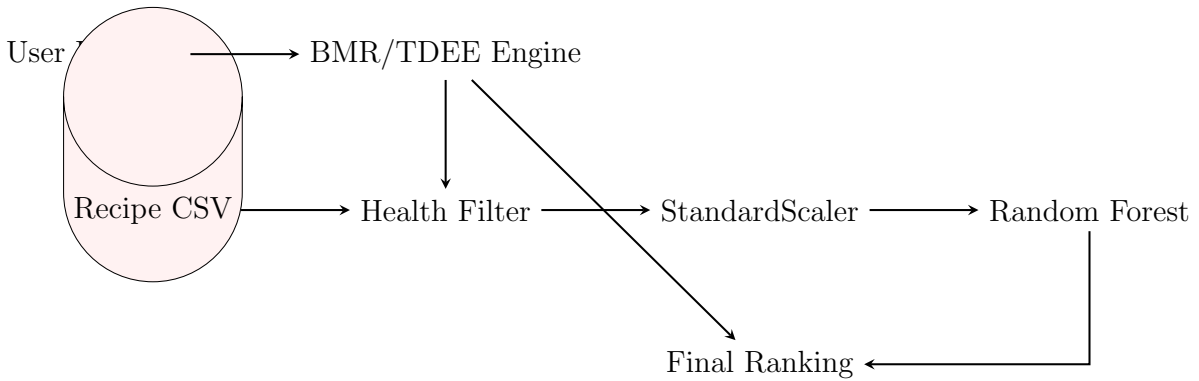


Figure 2: Low-Level Machine Learning and Filtering Pipeline

The low-level logic, detailed in Figure 2, operates as follows:

1. **Heuristic Calculations:** The system first determines clinical metrics like BMI and caloric needs based on activity multipliers.
2. **Constraint Satisfaction:** Recipes are strictly filtered from the 400+ row dataset based on medical compatibility (e.g., Diabetes or Thyroid-safe flags).
3. **Feature Inference:** Remaining candidates are scaled via `StandardScaler` and passed through the `RandomForestClassifier` to predict a recommendation score based on 28 trained features.
4. **Weighted Ranking:** A final score is calculated by balancing the AI prediction (50%), caloric proximity (20%), and goal compatibility (30%).

4 Implementation Challenges

The development process involved several technical hurdles that required iterative debugging and system optimization:

- **Data Preprocessing and Quality Control:** The raw dataset contained over 12,000 recipe entries. We implemented a quality filter to remove "false" values such as industrial supplements (e.g., protein isolates) and identified junk keywords like "gum" or "isolate" to ensure only whole-food recipes remained and the data set was reduced to around 400 rows.
- **Cold-Start Logic for Rare Dietary Intersections:** A significant challenge occurred when users presented highly restrictive overlapping profiles (e.g., a Vegan with Kidney Disease pursuing Muscle Gain). Since the finite 400-row dataset occasionally lacked an exact match for such niche intersections, the system would initially return an empty set. We resolved this by implementing a "Relaxed Constraint" fallback mechanism that prioritized medical safety flags while slightly broadening caloric tolerance to identify the most viable nutritional alternative.
- **Feature Importance Skewing in the Predictive Model:** During early testing of the Random Forest model, we observed a "Self-Correction Bias" where the algorithm prioritized recipe popularity (rating) over macronutrient density. This resulted in suboptimal recommendations for specific goals, such as suggesting high-rated but low-protein snacks for "Muscle Gain" profiles. To fix this, we performed a weighted feature recalibration during the training phase (`train_and_save_model`), manually increasing the coefficients for protein and goal-specific scores to ensure AI outputs aligned with the BMR engine's clinical targets.